Supporting Information for
BioInteract: Interpretable Drug–Target Interaction Prediction via Residue-Level Cross-Attention with Protein-Language and Physicochemical Priors

This document reports values recoverable from the released Davis records, configuration files, saved result files, source code, and the fixed BindingDB external-validation manifest included with this revision. "Classifier score" denotes the sigmoid output for the binary high-affinity class; it is not an experimental affinity. Model-native attention and Grad-CAM values are attribution scores, not contact assignments.

Model configuration and input features

Table S1: Model architecture confirmed from the checkpoint-embedded model configurations.

Component	Archived setting
Drug encoder	GINE; 3 layers; hidden dimension 256; dropout 0.20
Atom and bond input	52 node features; 16 edge features
Protein language model	`esm2_t30_150M_UR50D`; 640-dimensional residues
Protein projection	256-dimensional projection; 32-dimensional domain-label embedding
Cross-attention	8 heads; attention dimension 256; dropout 0.10
Pooling	Gated
Prediction head	Hidden dimensions 256 and 128; dropout 0.30; classification task

Table S2: Protein-residue input channels and their availability.

Channel	Archived implementation and limitation
ESM-2	Frozen 640-dimensional residue embeddings; proteins are truncated at 1,200 residues.
Physicochemical vector	Four values per residue: hydrophobicity, normalized molecular weight, charge, and polarity.
Domain-label channel	The configuration reserves a 32-dimensional embedding. The label builder uses `NONE` when no per-protein JSON annotation is available.
Annotation availability	No `data/domain_annotations/*.json` files are present in the released package. Thus the reproducible Davis input uses the unknown label at every residue and is not curated Pfam/InterPro information.

Table S3: Training provenance for the historical entity-split checkpoints and the separately trained exact-sequence-grouped strict models.

Setting	Value
Historical entity-split scope	Model architecture only; the three historical entity-split checkpoints do not retain optimizer, scheduler, warmup, or loss state
Historical Random and Drug-ID logs	Final-run headers record weighted BCE, label smoothing 0.05, AdamW (10^{-4}, 10^{-5}), and cosine warmup
Historical Target-ID trace	Older `run_split.py` header; only loss and validation AUROC are logged, so optimizer, warmup, and smoothing are not recoverable from the artifact
Historical threshold protocol	A validation F1 threshold is frozen before each canonical test evaluation
Strict-model initialisation and data	Each exact-sequence-grouped model was newly initialised with seed 42 and trained only on its new Davis training partition, using the same preprocessing and architecture
Strict-model optimisation	Weighted BCE with 0.05 label smoothing; AdamW (learning rate 10^{-4}, weight decay 10^{-5}); batch size 64; gradient-norm clipping 1.0; five-epoch linear warm-up followed by cosine decay; maximum 100 epochs
Strict-model selection	Highest validation AUROC with patience 15; validation F1 threshold selected after loading that checkpoint and frozen for test evaluation
Strict cold-target artifact	`revision/analysis/sequence_grouped_cold_target.pt`; best epoch 24 of 39; validation AUROC 0.911568; threshold 0.837581; SHA-256 `8ea58e1f056b5e33c309787880a811c797393510c459d9442b5dbe67e632fe5c`
Strict cold-both artifact	`revision/analysis/sequence_grouped_cold_both.pt`; best epoch 6 of 21; validation AUROC 0.910630; threshold 0.935704; SHA-256 `8b7c0f374983a35163a7d3243c89732504a5a97ef4d10c557b2841a627873972`

Table S4: Computational environment documented during final reproducibility audit.

Item	Value
Python	3.10.14
PyTorch	2.4.0+cu121
PyTorch Geometric	2.6.1
RDKit	2025.3.6
`fair-esm`	2.0.0
Biopython and RapidFuzz	1.87 and 3.14.5
CUDA availability during audit	Available
Audit GPU	NVIDIA GeForce RTX 4080
Strict-split run record	`device: cuda` in `strict_split_metrics.json`
Historical checkpoint files	`best.pt`, `best_random.pt`, `best_cold_target.pt`, `best_cold_drug.pt`
Strict-model checkpoint files	`revision/analysis/sequence_grouped_cold_target.pt` and `revision/analysis/sequence_grouped_cold_both.pt` (hashes in Table S3)

Dataset, partitions, and metrics

Table S5: Canonical current-release Davis test-partition summaries. Values are from the full-precision checkpoint reconstruction, frozen validation thresholds, and the associated prediction files.

Partition	Test pairs	Positives	AUROC	AUPRC	F1	Precision	Recall
Random	6,011	276	0.904	0.560	0.565	0.602	0.533
Target-ID-held-out	5,984	250	0.930	0.525	0.534	0.517	0.552
Drug-ID-held-out	5,746	245	0.733	0.167	0.100	0.273	0.061

Table S6: Bootstrap intervals for the canonical current-release entity-based partitions. Each interval uses 600 ordinary pair-level bootstrap replicates sampled with replacement at the test-pair level, NumPy seed 42, and the 2.5th and 97.5th percentiles of valid replicate metrics. Replicates containing only one class are discarded. The main figure shows point estimates only.

Partition	AUROC 95% interval	AUPRC 95% interval
Random	[0.881, 0.924]	[0.498, 0.624]
Target-ID-held-out	[0.911, 0.948]	[0.454, 0.584]
Drug-ID-held-out	[0.703, 0.763]	[0.129, 0.211]

Table S7: Exact-sequence-grouped strict within-Davis evaluations from `strict_split_metrics.json`. The two models were trained from initialisation under the protocol in Table S3. Threshold-dependent quantities use the corresponding validation-set threshold; all strict-split entries are point estimates, not bootstrap confidence intervals.

Pair accounting

Protocol	Train pairs	Validation pairs	Test pairs	Test positives	Excluded cross-block pairs
Exact-sequence-grouped cold-target	21,080	2,992	5,984	208	0
Exact-sequence-grouped cold-both	15,190	264	1,144	38	13,458

Test metrics

Protocol	AUROC	AUPRC	Threshold	F1	Precision	Recall
Exact-sequence-grouped cold-target	0.873	0.383	0.838	0.468	0.517	0.428
Exact-sequence-grouped cold-both	0.552	0.038	0.936	0.000	0.000	0.000

For the cold-both protocol, drug identifiers and exact target-sequence groups are partitioned independently at 70/10/20. Only pairs in the corresponding train–train, validation–validation, and test–test blocks are retained; the remaining cross-block pairs are excluded. The cold-both validation block contains 10 positive pairs.

Reproducibility details for revision analyses

Identifier-partition similarity audits. Drug similarities use RDKit 2025.3.6 Morgan fingerprints (radius 2, 1,024 bits, `includeChirality=False`) and Tanimoto similarity. Protein similarities for the original Target-ID-held-out audit use RapidFuzz 3.14.5 `fuzz.ratio`/100. For sequences x, y, this is the normalised Indel similarity $1 - d_{\text{Indel}}/(|x| + |y|)$, where substitutions are represented as one deletion plus one insertion.

Strict-test global alignment. Biopython 1.87 `Align.PairwiseAligner` is used in global mode with match = 1, mismatch = 0, gap-opening = -1, and gap-extension = -0.5. Full-length identity is the number of exact-residue matches divided by alignment length; the maximum across all training targets is retained for each held-out target.

Bootstrap intervals. The saved 95% intervals use ordinary, unstratified pair-level resampling with replacement, 600 requested replicates, and NumPy random seed 42. Each replicate contains the same number of test pairs as the corresponding test set; one-class replicates are skipped. Bounds are the 2.5th and 97.5th percentiles of the valid AUROC or AUPRC replicate distribution.

Grad-CAM. Hooks are attached to `drug_encoder.layers.2`, the third and final GINE message-passing layer. Gradients are taken from the pre-sigmoid positive-class logit, pooled across graph nodes to obtain channel weights, rectified, and then normalised separately within each molecular graph.

Attention sparsity. For pair q, residue scores $s_{qj} = \sum_i M_{qij}$ are normalised as $\widetilde{s}_{qj} = s_{qj}/\max_k s_{qk}$. The reported fraction is $|\{(q,j) : \widetilde{s}_{qj} < 0.1\}|/\sum_q L_q$. It uses all 1,506 positive Davis pairs with the archived `checkpoints/best.pt`, not a single test set. Every pair is normalised before the 1,301,380 valid residue scores are concatenated; thus original training, validation, and test memberships are all included and longer targets contribute more residue-level observations. The archived flat-score array is `BioInteract/results/figure_data/attention_distribution.npz`; it is the sole source of the attention-distribution figure and cumulative curve in the main text.

Residue-attribution summary.

Quantity	Value
Positive Davis pairs	1,506
Residue-score observations	1,301,380
Mean / standard deviation	0.004351 / 0.046243
Median	0.000272
95th / 99th percentile	0.003617 / 0.057193
Fraction below 0.1	0.992319
Attribution mass in top 1% / 5% scores	0.788528 / 0.904465

Exact-sequence-grouped target diagnostic controls. All values use the 5,984-pair strict target test set with 208 positives.

Control	AUROC	AUPRC
BioInteract	0.873	0.383
Drug-only Morgan logistic regression	0.764	0.164
Drug-promiscuity prior	0.764	0.163
Target-only constant prior	0.500	0.035
Label-shuffled drug-only control	0.407	0.029

Feature and input audits

Table S8: Atom and bond feature specification from `src/data/mol_graph.py`.

Level	Feature family	Dimensions
Atom	Atom type one-hot (20 choices including unknown)	20
Atom	Hybridization, formal charge, and hydrogen-count one-hot vectors	18
Atom	Degree, aromaticity, and ring membership	3
Atom	Ring-size membership for sizes 3–8	6
Atom	Donor/acceptor, Crippen logP and MR, Gasteiger charge	5
Atom total		52
Bond	Bond type, conjugation, ring membership, stereo, and direction	16

Table S9: Davis nominal ABL1 input audit. Every listed record has the same complete SHA-256 digest `cd6a78b82fede215cdb80a797f228098a41178fdde0753728f3781e6221fb1ee`; values in the last columns are the archived D0010 and D0017 affinities in nM. The nominal labels do not alter the model sequence input.

Target ID	Nominal name	Length	Matches ABL1	D0010	D0017
T0001	ABL1(E255K)	1167	Yes	0.047	0.047
T0002	ABL1(F317I)	1167	Yes	0.63	0.10
T0003	ABL1(F317I)p	1167	Yes	0.18	0.041
T0004	ABL1(F317L)	1167	Yes	0.11	0.032
T0005	ABL1(F317L)p	1167	Yes	0.029	0.019
T0006	ABL1(H396P)	1167	Yes	0.062	0.025
T0007	ABL1(H396P)p	1167	Yes	0.057	0.046
T0008	ABL1(M351T)	1167	Yes	0.037	0.016
T0009	ABL1(Q252H)	1167	Yes	0.086	0.037
T0010	ABL1(Q252H)p	1167	Yes	0.039	0.064
T0011	ABL1(T315I)	1167	Yes	21	890
T0012	ABL1(T315I)p	1167	Yes	3.6	120
T0013	ABL1(Y253F)	1167	Yes	0.036	0.058
T0014	ABL1	1167	Yes	0.12	0.029
T0015	ABL1p	1167	Yes	0.057	0.046

Table S10: Saved non-mutant EGFR–D0014 functional-group attribution values.

Functional group	Attribution
Carbonyl	0.431
Amide	0.389
Ether	0.136
Halogen	0.134
Amino	0.090
Aromatic ring	0.000
Heterocycle N	0.000

Table S11: Saved non-mutant BRAF–D0023 failure-mode summary. The archived affinity is positive at the study threshold, whereas the classifier score is low. The score and attributions are model outputs, not an explanation of binding.

Quantity	Value
High-affinity-class score	0.099
Archived Davis affinity (nM)	0.19
Aromatic-ring attribution	0.059
Hydroxyl attribution	0.000
Heterocycle-N attribution	0.000

Table S12: Trainable parameter count recovered from `interpretability_report.json`.

Quantity	Value
Total trainable parameters	2,442,083

Strict external BindingDB evaluation

Table S13 records the frozen external evaluation requested by the reviewers. The official curated-articles snapshot was handled independently of Davis, then filtered under the pre-specified exact-K_d and exact double-novel protocol. This is an external exact-entity test, not a claim of scaffold or remote-homology novelty, and no BindingDB label was used to retrain, calibrate, select the checkpoint, or select the threshold.

Table S13: Strict BindingDB external-validation provenance, curation, and frozen result. The source archive was `BindingDB_BindingDB_Articles_202607_tsv.zip` (SHA-256 `d2584d1519318d00ab5f46289da5ab3549affe732d598a5072f8777b6b3b5262`). Only finite exact K_d measurements in nM for valid single-chain proteins (maximum 1,200 residues) and valid ligand structures were considered. Exact canonical-isomeric-SMILES and full-normalised-sequence matches to every Davis ligand or target were excluded. Eligible measurements were grouped by canonical ligand–sequence pair; threshold-conflicting groups were removed at the pair level, and every remaining pair retained its median exact K_d. Confidence intervals are pair-stratified bootstrap 95% intervals from 600 replicates (seed 42).

Item	Value
Source snapshot	`BindingDB_BindingDB_Articles_202607` (official BindingDB curated articles TSV)
Source rows	93,712
Final positives / negatives	516 / 1,416
Unique ligands / targets	1,195 / 396
Sequential row-level exclusions	82,988 censored/non-exact K_d; 8,261 multi-chain; 139 invalid sequences; 8 invalid SMILES; 64 Davis-ligand matches; 94 Davis-sequence matches
Eligible measurements before pair reconciliation	2,158
Threshold-conflicting replicate groups	15 ligand–target pairs, comprising 44 measurement rows; conflicts are counted at the pair level
Threshold-consistent measurement rows	2,114
Median replicate aggregation	143 retained pairs had $\geq$ 2 consistent measurements; 182 repeated measurement rows were collapsed by the within-pair median K_d
Final external pairs	1,932
Frozen model and threshold	`checkpoints/best_random.pt`; 0.5959881544 (existing Davis random-validation threshold)
Protein features	Newly extracted `fair-esm` 2.0.0 `esm2_t30_150M_UR50D`; all 396 caches passed 640-dimensional and length checks
AUROC (95% CI)	0.5597 [0.5313, 0.5885]
AUPRC (95% CI)	0.3404 [0.3138, 0.3724]
Frozen-threshold F1 / precision / recall	0.0701 / 0.7308 / 0.0368
Two-run reproducibility	Identical per-pair probabilities; the complete SHA-256 is recorded in `bindingdb_external_manifest.json`
Interpretation	Limited external transfer; the result does not support a broad DTI-transfer claim and is not used for structural or attribution interpretation.

Breadth check across model families

The main-text comparison in Table 1 is a single-variable ladder: the same molecular graph encoder throughout, first without a protein language model, then with one, then with cross-attention in place of concatenation. Table S14 places that ladder alongside two further architectures trained by the same harness, under the same protocol described in Table S16, on the random split. The two additional models are a sequence convolutional model in the DeepDTA family and the low-rank bilinear attention and bilinear pooling of DrugBAN applied to the same two encoders used by BioInteract. They were run on the random split only, as a breadth check across families rather than as a second controlled contrast, and their protocol-dependence was therefore not assessed.

The five architectures fall within roughly 0.03 AUROC of one another on this split: the sequence convolutional model, which uses neither a molecular graph nor a protein language model, reaches AUROC 0.912 and AUPRC 0.569, the bilinear interaction module reaches 0.907 and 0.547, and BioInteract reaches 0.913 and 0.545. This is the basis for the statement in the main text that the random split separates architectures weakly, so that a favourable random-split number is poor evidence for a design choice and the informative contrasts are the entity-held-out ones.

Table S14: Random-split breadth check. All values were measured in this study under the shared protocol of Table S16. The first three rows are the main-text ladder; the last two are the additional architectures. "PLM" indicates whether the model consumes the frozen `esm2_t30_150M_UR50D` residue representation. Best value in each column is in bold.

Model	Interaction paradigm	PLM	AUROC	AUPRC
GraphDTA	Graph + sequence CNN	No	0.885	0.358
ESM2-GraphConcat	Graph + PLM, concatenation	Yes	0.907	0.478
BioInteract (ours)	Graph + PLM, cross-attention	Yes	**0.913**	0.545
DeepDTA	Sequence CNN	No	0.912	**0.569**
ESM2-Bilinear	Graph + PLM, bilinear attention	Yes	0.907	0.547

Identity-stratified cold-target results

Table S15 gives the numerical values plotted in main-text Figure 3. Held-out targets in the exact-sequence-grouped cold-target partition are grouped by their maximum full-length global sequence identity to any training target, computed with the Biopython global aligner (match = 1, mismatch = 0, gap opening = -1, gap extension = -0.5); identity is exact-residue matches divided by full alignment length. The two outer strata contain 75 and 3 positive pairs respectively, so their AUPRC estimates are unstable; the ordering across strata is not monotone and should not be read as a homology-transfer trend.

Table S15: BioInteract performance on the exact-sequence-grouped cold-target test, stratified by maximum global sequence identity to the training targets. The positive prevalence of the full cold-target test is 0.035.

Maximum identity	Test targets	Test pairs	Positive pairs	Positive prevalence	AUROC	AUPRC
< 0.40	31	2,108	75	0.036	0.806	0.305
0.40–< 0.60	28	1,904	77	0.040	0.908	0.456
0.60–< 0.80	24	1,632	53	0.032	0.908	0.466
≥ 0.80	5	340	3	0.009	0.862	0.121
All	88	5,984	208	0.035	0.873	0.383

Matched-protocol comparison provenance

Table S16 records one row per training run behind main-text Table 1. Every run used the same Davis records, the same partition for its protocol, seed 42, weighted binary cross-entropy with 0.05 label smoothing, AdamW (learning rate 10^{-4}, weight decay 10^{-5}), an effective batch size of 64 assembled from micro-batches of 16 with gradient accumulation, automatic mixed precision, gradient-norm clipping at 1.0, a five-epoch linear warm-up followed by cosine decay, a maximum of 30 epochs, and early stopping at patience 6 on validation AUROC. Validation and test inference were run in full precision. The reported threshold was selected once on that run's validation predictions by maximising F1 and then frozen for the test evaluation. Each configuration was trained once; repeated-seed variability was not estimated, so small differences between adjacent rows should not be interpreted.

Table S16: Per-run provenance for the matched-protocol comparison. "Best epoch" is the epoch whose validation AUROC was highest and whose weights were restored before the test evaluation.

Model	Protocol	Parameters	Epochs	Best epoch	Val. AUROC	Threshold	Test AUROC	Test AUPRC
BioInteract	random	2,442,083	21	15	0.9456	0.9036	0.9128	0.5454
DeepDTA	random	919,169	30	27	0.9316	0.9207	0.9125	0.5692
ESM2-Bilinear	random	1,533,697	30	28	0.9175	0.8270	0.9065	0.5470
ESM2-GraphConcat	random	1,335,553	30	29	0.9236	0.8277	0.9071	0.4785
GraphDTA	random	2,264,513	26	20	0.8895	0.8822	0.8851	0.3581
BioInteract	target_id_held_out	2,442,083	25	19	0.9214	0.8867	0.9340	0.5693
DeepDTA	target_id_held_out	919,169	30	26	0.8846	0.7149	0.9136	0.4932
ESM2-GraphConcat	target_id_held_out	1,335,553	23	17	0.8553	0.8958	0.8780	0.3687
GraphDTA	target_id_held_out	2,264,513	22	16	0.8457	0.6843	0.8557	0.3056
BioInteract	drug_id_held_out	2,442,083	7	1	0.5708	0.6939	0.7126	0.1259
ESM2-GraphConcat	drug_id_held_out	1,335,553	7	1	0.5833	0.5038	0.6836	0.0968
GraphDTA	drug_id_held_out	2,264,513	7	1	0.6180	0.6050	0.6793	0.1050

BindingDB filtering waterfall and snapshot provenance

Table S17 records the provenance of the two BindingDB snapshots audited in this revision and the reconciliation of the two filter orderings. Both orderings apply exactly the same criteria and both retain the same number of eligible measurements for a given snapshot; they differ only in the order in which the measurement-quality and target-validity tests are applied, which changes how individual rows are attributed between the first three counters. The measurement-first ordering used in main-text Table 4 was adopted because it isolates the dominant cause of the reduction, namely rows that report no K_d at all.

Table S17: Provenance of the two BindingDB snapshots and reconciliation of the two audit orderings. The curated-articles snapshot is the pre-specified source of the frozen external evaluation; the complete release was added in this revision as an independent verification that the filter definitions scale.

Item	Curated articles	Complete release
Source version	`BindingDB_BindingDB_Articles_202607`	`BindingDB_All_202608`
Archive file	`BindingDB_BindingDB_Articles_202607_tsv.zip`	`BindingDB_All_202608_tsv.zip`
Archive bytes	18,114,757	592,986,963
Archive SHA-256	`d2584d1519318d00ab5f46289da5ab35 49affe732d598a5072f8777b6b3b5262`	`8dc30541d668e5e403ba1fe9a682b76c 42a50e1711aac44329c62ba24ccd57db`
TSV member	`BindingDB_BindingDB_Articles.tsv`	`BindingDB_All.tsv`
Input rows	93,712	3,234,499
Rows with no K_d reported	91,062 (97.2%)	3,104,331 (96.0%)
Eligible measurements, measurement-first ordering	2,158	52,085
Eligible measurements, single-chain-first ordering	2,158	52,085
Pair-level threshold conflicts removed	15 pairs / 44 rows	1,992 pairs / 5,233 rows
Retained double-novel pairs	1,932	38,193
Retained positives / negatives	516 / 1,416	13,264 / 24,929
Unique ligands / targets	1,195 / 396	26,433 / 3,462
Frozen AUROC (95% CI)	0.560 [0.531, 0.589]	0.533 [0.526, 0.539]
Frozen AUPRC (95% CI)	0.340 [0.314, 0.372]	0.383 [0.378, 0.390]
Frozen-threshold F1 / precision / recall	0.070 / 0.731 / 0.037	0.044 / 0.513 / 0.023
Prediction SHA-256	`9e50dfadbdcb9974d6bc2dcefbdc4edb f4bd089b14598d76724da48fa40bba1c`	`303ba15ab0e605607102d08e7b2fc3c4 cee888d11cc531d5aced121d00cf9013`
Repeat-run check	Two runs gave identical per-pair probabilities	Evaluated once; the repeat run was not performed
Role in this study	Pre-specified frozen external evaluation	Verification of filter scaling and a larger frozen external evaluation